

o Aim:

- To study the transient response of a series RC circuit and understand the time constant concept using waveforms.

o Apparatus:

Equipment	Range	Quantity
Function Generator	0.1 Hz - 100 MHz	1
Digital storage Oscilloscope (DSO)	-	1
Decade Resistance Box (DRB)	10 k Ω	1
Decade Capacitance Box (DCB)	220 nF	1
Breadboard	-	1
Connecting wires	-	Required

o Theory:

- In this experiment, we apply a pulse waveform to the RC circuit to analyze the transient response of the circuit. The pulse-width relative to a circuit's time constant determines how it affects an RC circuit.
- Time constant (τ): A measure of time required for certain changes in voltages and currents in RC and RL circuits. Generally, when the time elapsed exceeds five time constants (5τ) after switching has occurred, the currents and voltages have reached their final value, which is also called 'steady-state response'. The time constant of an RC circuit is the product of equivalent capacitance and the Thevenin's resistance as viewed from the terminals of equivalent capacitor.

$$\therefore \tau = RC \quad \text{--- (1)}$$

o Circuit Diagrams:

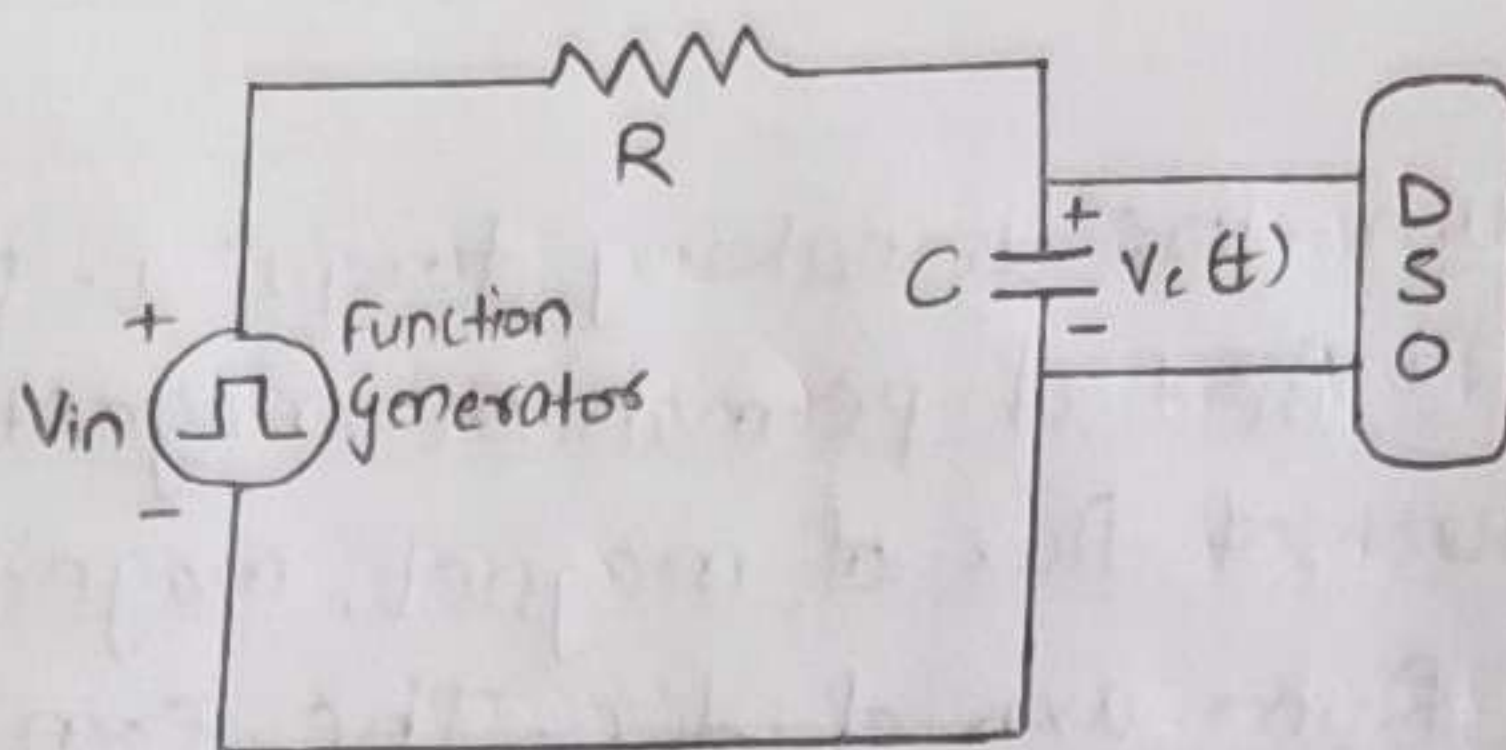


Figure (1)

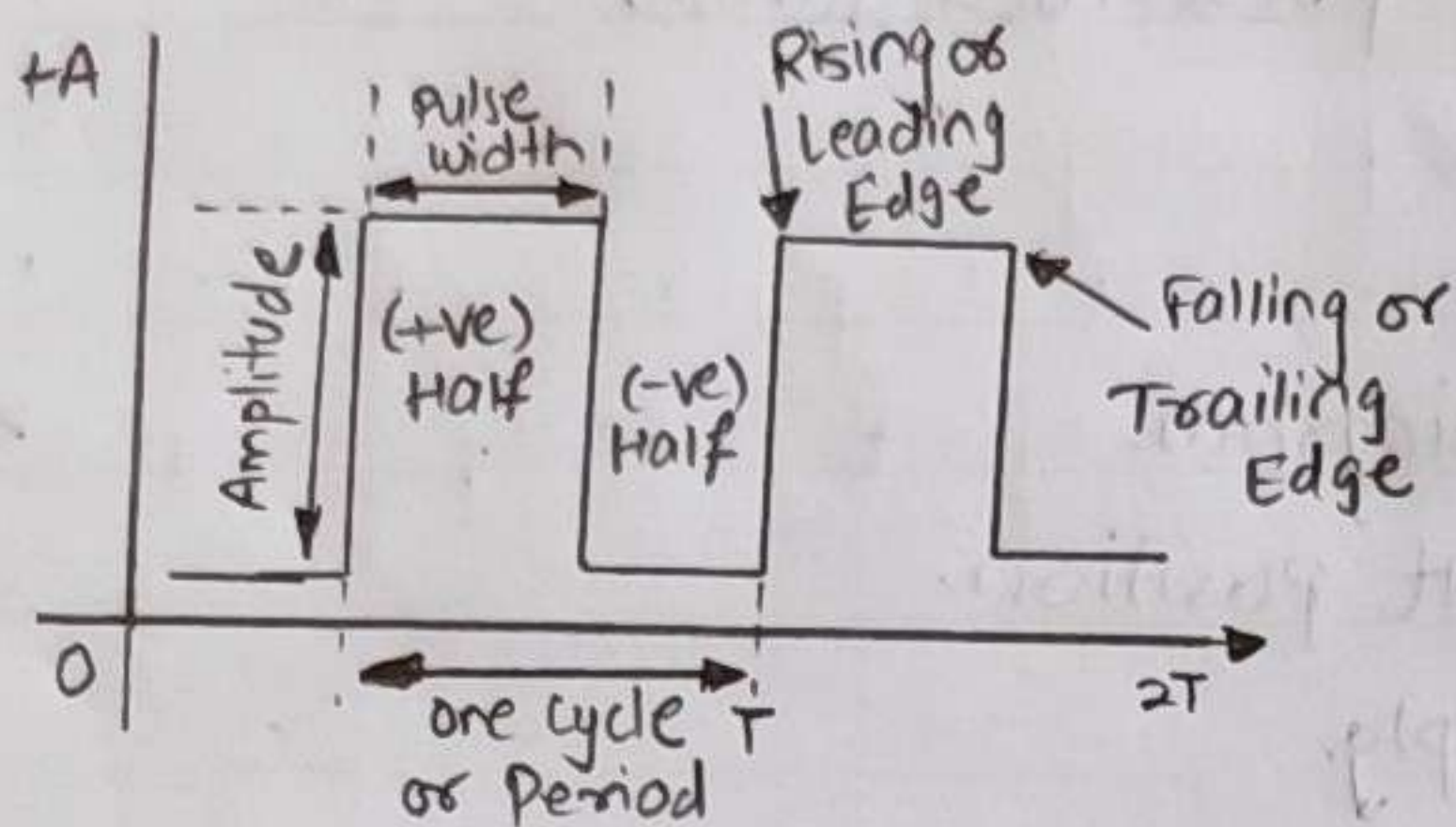


Figure (2)

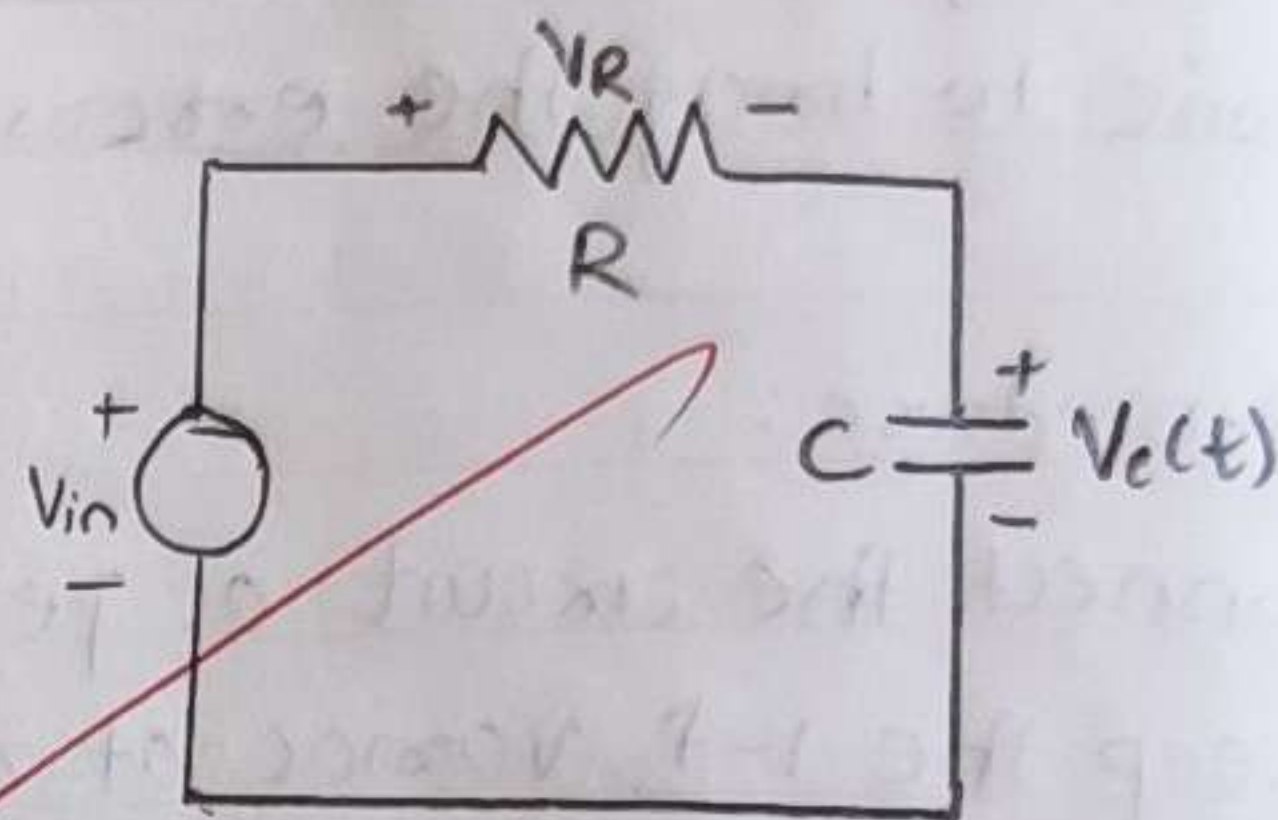
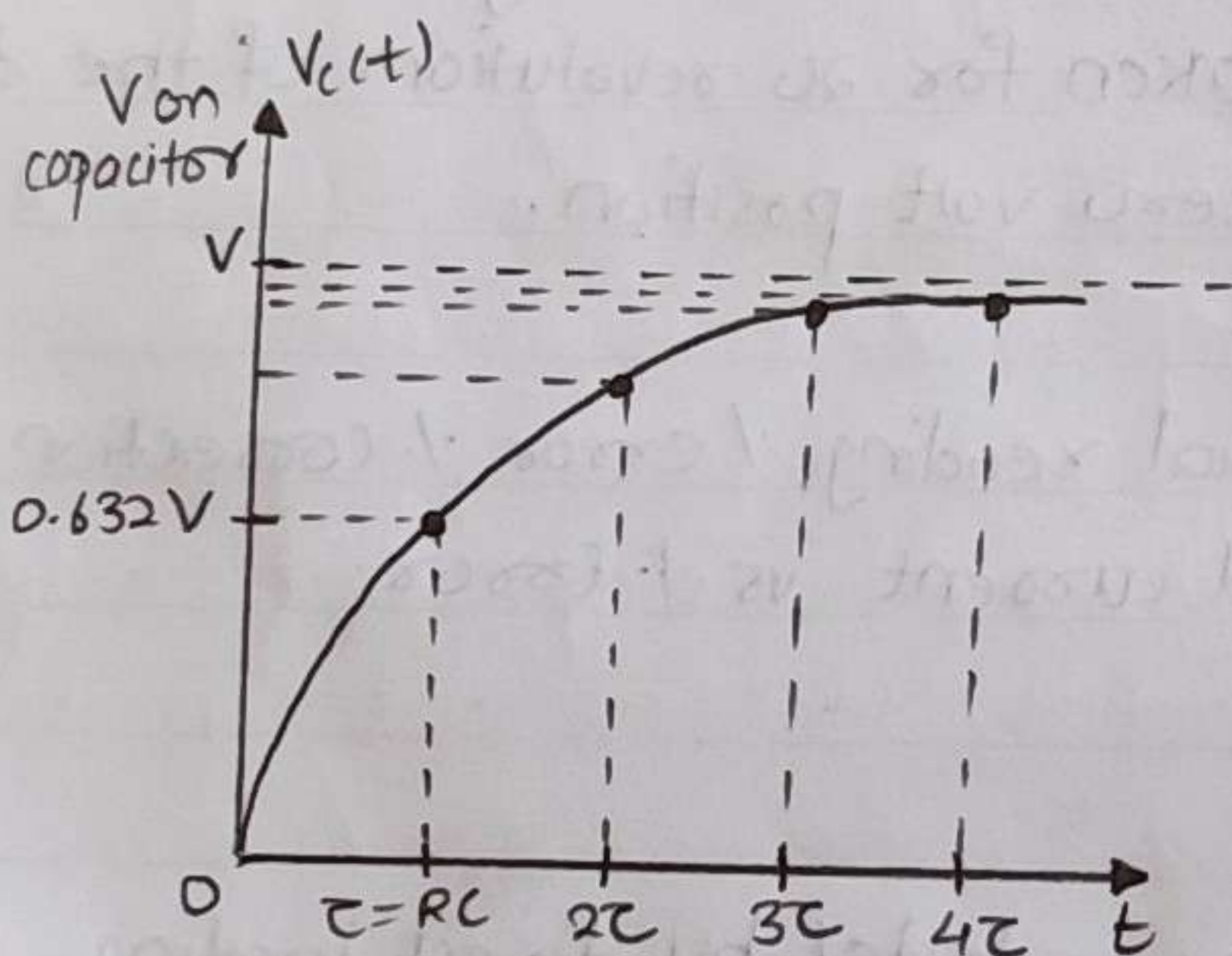
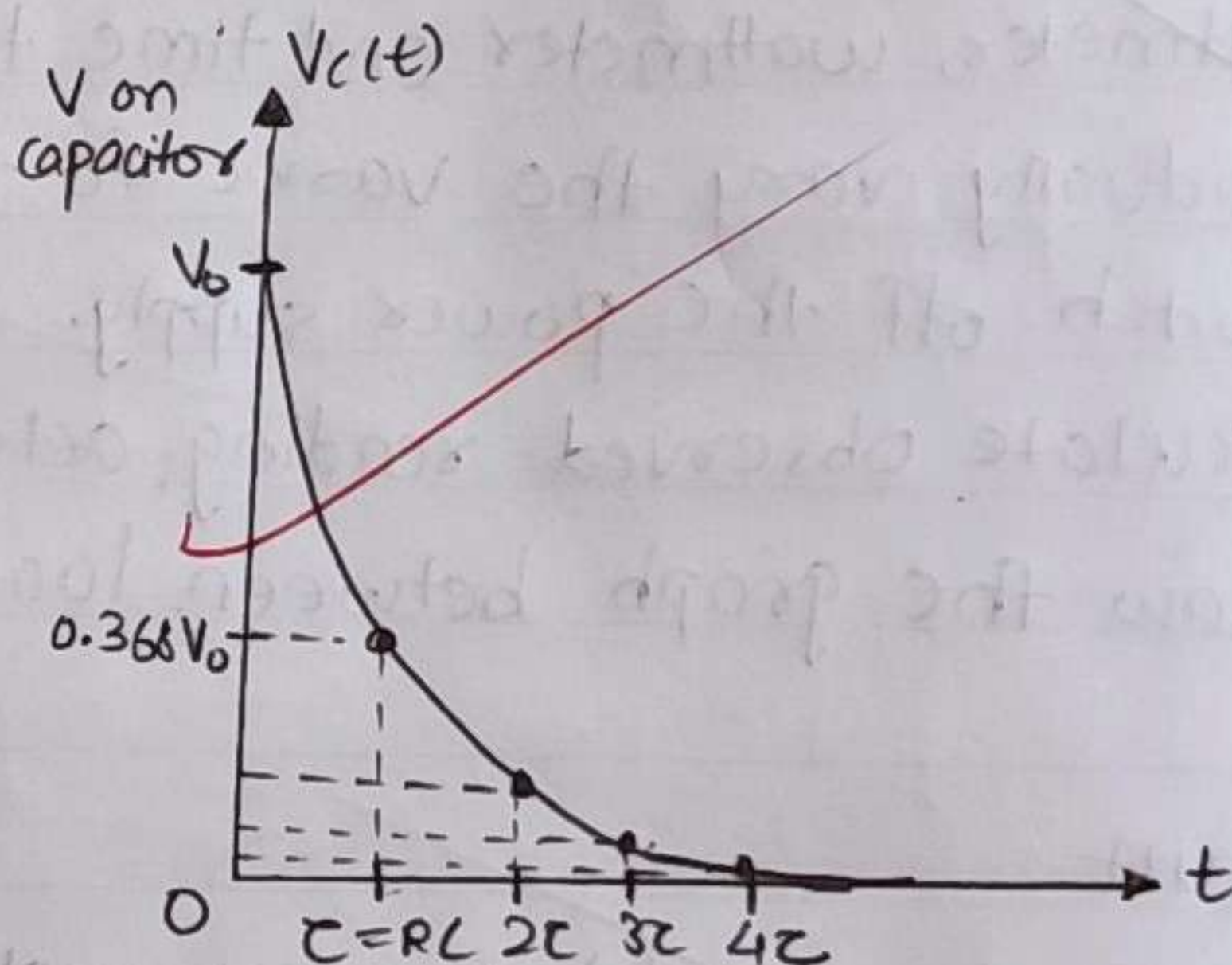


Figure (3)



charging of capacitor
 $V_c(t) = V(1 - e^{-t/RC})$

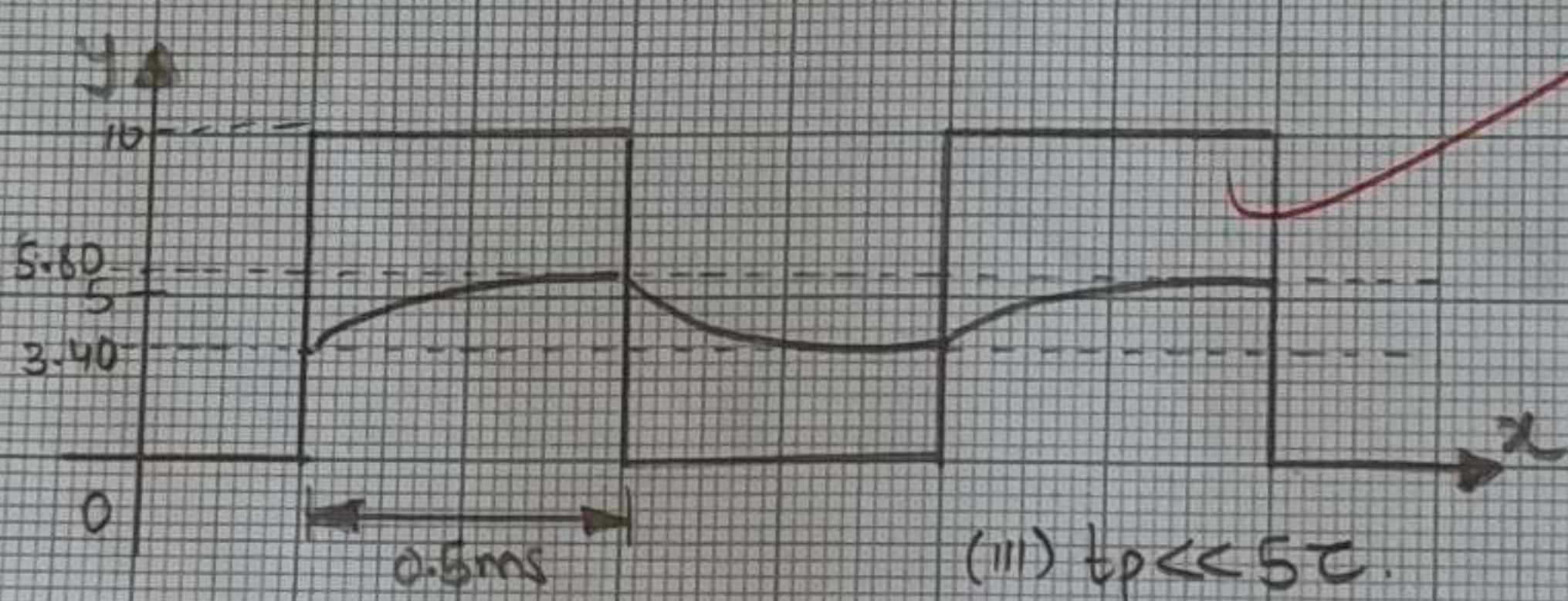
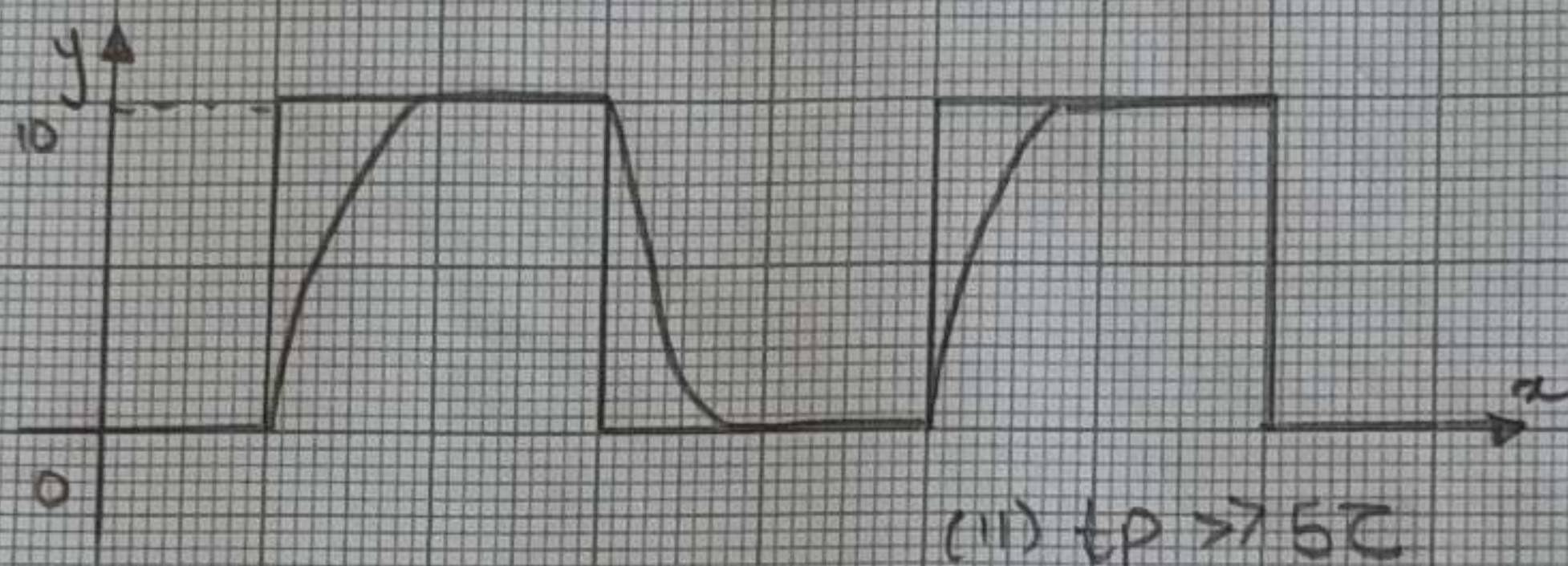
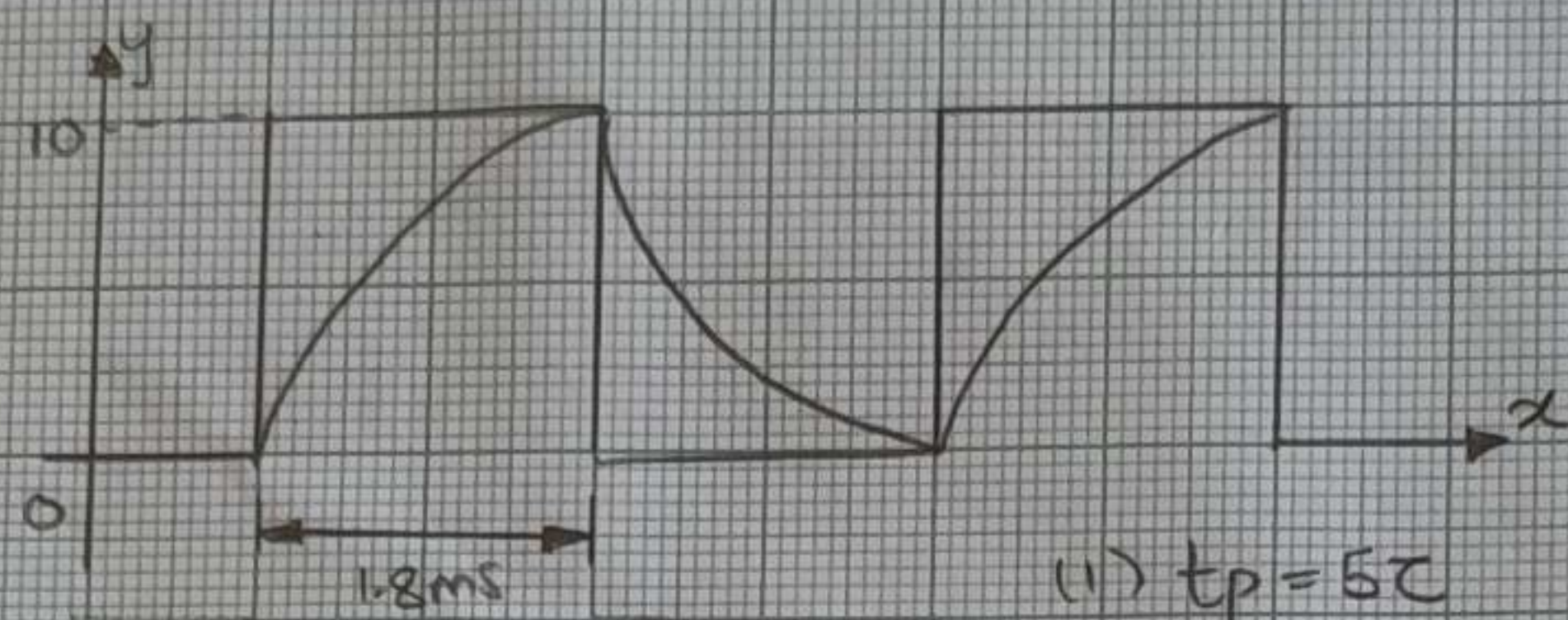


Discharging of capacitor
 $V_c(t) = V_0 e^{-t/RC}$

o Observation Table:

Cases	Time period $t_p(\text{ms})$	Theoretical Time Constant ($\tau = RC$)	Practical Time Constant
1) $t_p = 5\tau$	11	2.2 ms	1.8 ms
2) $t_p \gg 5\tau$	110	2.2 ms	2.24 ms
3) $t_p \ll 5\tau$	1	2.2 ms	0.5 ms

o Graphs:



~~Very
2.2 ms~~

- A pulse is a voltage or current that changes from one level to the other and back again. If a waveform's high time is equal to its low time, as in figure (2), it is called as a 'square wave'. The length of each cycle of a pulse train is termed as its period (T). The pulse width (t_p) of an ideal wave is equal to half the time period.

- The relation b/w pulse width & frequency is: $f = \frac{1}{2t_p}$ - (ii).

- Consider figure (3). From Kirchhoff's laws, it shows that the charging voltage $V_c(t)$ across the capacitor is given by:

$$V_c(t) = V(1 - e^{-t/RC}), \quad t \geq 0 \quad \text{--- (iii)}$$

where V is the applied source voltage to the circuit for $t = 0$.
 $RC = \tau = \text{time constant}$.

- The discharging voltage for the capacitor is given by:

$$V_c(t) = V_0 e^{-t/RC}, \quad t \geq 0 \quad \text{--- (iv)}$$

where V_0 is the initial voltage stored in the capacitor at $t = 0$.
 $RC = \tau = \text{time constant}$.

• Procedure:

- Make the connections as per the circuit diagram given.
- From the function generator, apply 4V p-p square wave as input.
- Observe the response of the circuit for the follⁿ three cases:
 - i) $t_p \gg 5\tau$: Set frequency of FGEN output such that the capacitor has enough time to fully charge and discharge during each cycle of the square wave. Let $t_p = 25\tau$ and accordingly set the FGEN frequency using eq (ii). Determine the time constant from the waveforms obtained on the DSO screen.

ii) $t_p = 5\tau$: since the pulse width is exactly 5τ , the capacitor should just be able to fully charge and discharge during each pulse cycle.

iii) $t_p < 5\tau$: In this case, the capacitor does not have time to charge significantly before it is switched to discharge, and vice versa.
Let $t_p = 0.5\tau$ and set the frequency accordingly.

- Repeat the steps for new values of R and C, and verify theoretically calculated and practically measured time constants.

° Result:

- Transient Response is observed and time constant is verified.

~~Temp
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